

The Evolution of Automated Fake News Detection

From Classical ML to Bidirectional LSTMs and BERT

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Why Does This Matter?

The fake news problem at scale

The Problem

- ▶ Misinformation spreads 6× faster than true news on social media
- ▶ 2016 US election: fake stories outpaced real news in Facebook engagement
- ▶ Public health crises amplified by vaccine and pandemic misinformation
- ▶ Manual fact-checking cannot scale to millions of posts per day
- ▶ Automated detection is no longer optional - it is essential

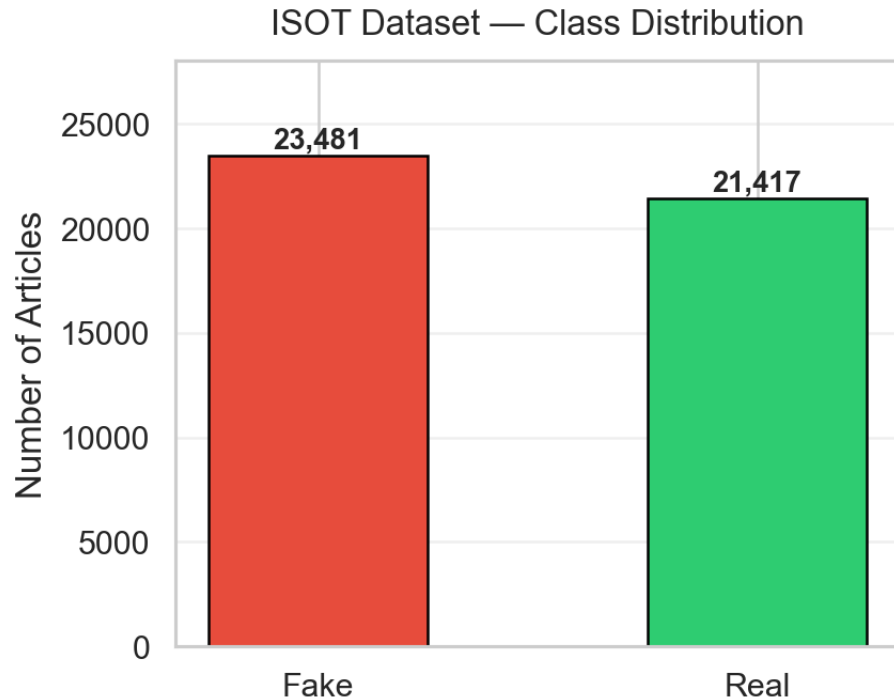
My Approach

- ▶ Systematically compare three generations of NLP methods
- ▶ Shared benchmark: ISOT Fake News Dataset (~45 000 articles)
- ▶ Fair evaluation under identical preprocessing conditions
- ▶ Document the accuracy vs. complexity trade-off

Classical ML → Deep RNN / LSTM → BERT

Dataset: ISOT Fake News Corpus

University of Victoria — publicly available on Kaggle



Dataset at a Glance

- ▶ 44 898 labelled news articles
- ▶ 21 417 real articles (Reuters, 2015–2018)
- ▶ 23 481 fake articles (Politifact-flagged sources)
- ▶ Fields: title, text, subject, date

Preprocessing Pipeline

- ▶ Lowercase → Remove URLs
- ▶ Strip non-alphabetic characters → Collapse whitespace
- ▶ NLTK stopwords removal (classical ML only)
- ▶ Vocabulary capped at 15 000 tokens (deep models)
- ▶ Sequences truncated / padded to 150 tokens

Three Generations of Detection Methods

Generation 1 Classical ML

- TF-IDF bag-of-words features
- Naive Bayes
- Logistic Regression
- Random Forest (100 trees)
- Full 36k training set

Generation 2 Deep Recurrent

- Learned word embeddings
- Vanilla RNN (vanishing ∇)
- Bidirectional LSTM (gates)
- 2.000-sample quick-mode run
- PyTorch from scratch

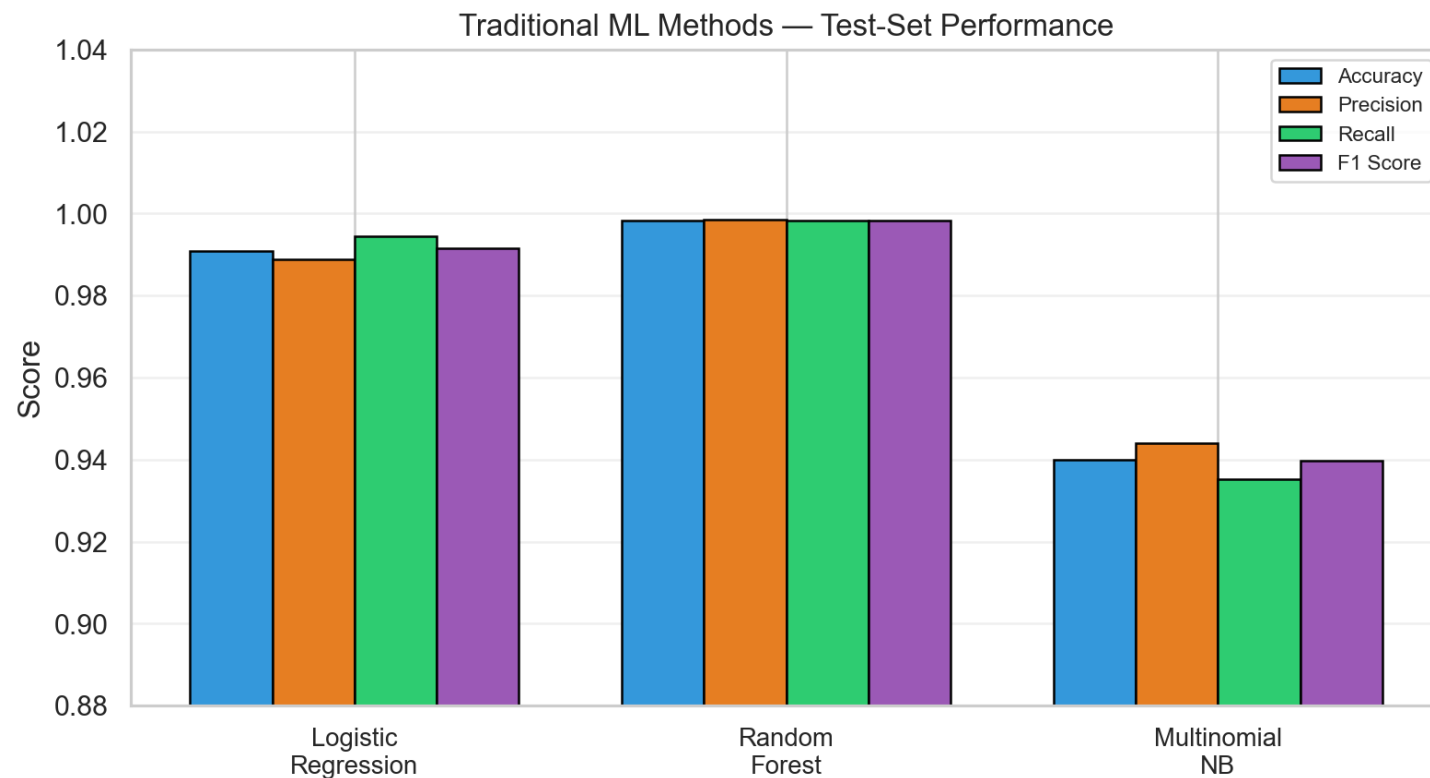
Generation 3 BERT Transformer

- Pre-trained on 3.3B words
- Self-attention (full context)
- WordPiece tokenisation
- 110M parameters fine-tuned
- Test Accuracy: 99.87% ✓

All models evaluated on the same test set

Classical ML Results

TF-IDF + Sklearn, 8 980-sample test set



Key Numbers

Naive Bayes Acc.

94.0 %

Log. Regression

99.1 %

Random Forest

99.8 %

RF mis-classified

15 / 8 980

Takeaway

Strong stylistic signals in the ISOT corpus let even a bag-of-words model exceed 94 %. Random Forest is near-perfect at 99.8 %, but has zero understanding of language semantics.

Vanilla RNN — Architecture & Results

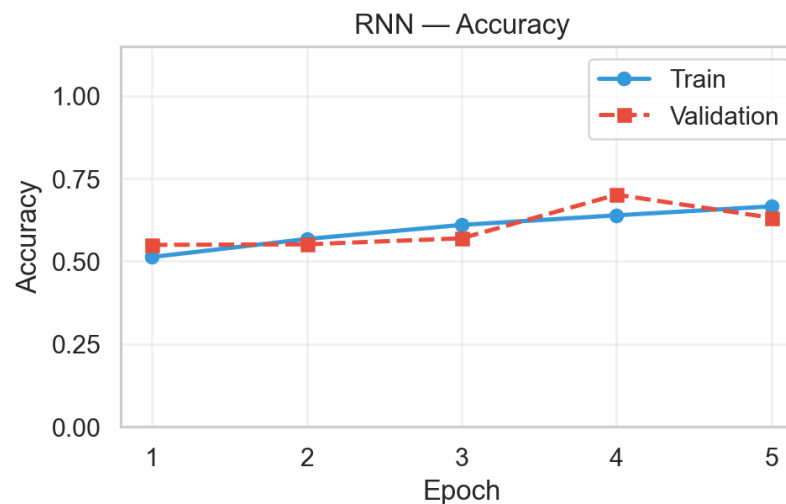
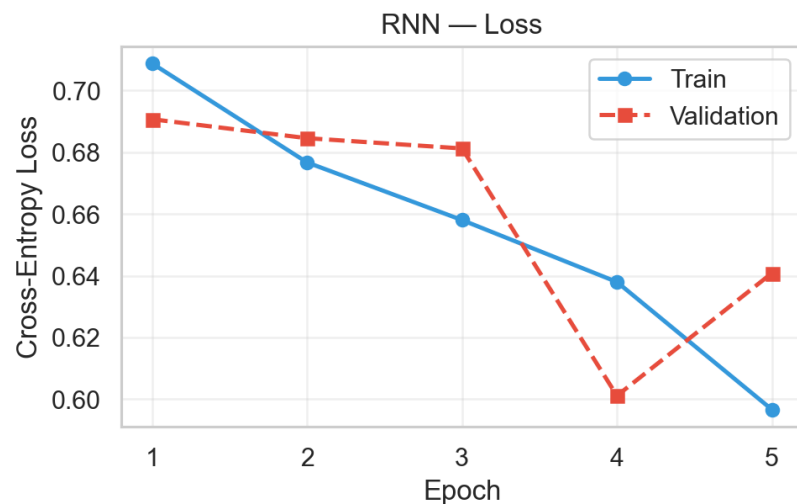
2 000 training samples · 5 epochs · CPU quick mode

How it works

$$h_t = \tanh(W_{hh} \cdot h_{t-1} + W_{xh} \cdot x_t + b)$$

- Processes tokens one at a time, left → right
- Hidden state carries memory of the whole prefix
- Vanishing gradients → forgets early tokens

RNN Training History (Quick Mode, 5 Epochs)



Test Results

Accuracy

70.3 %

F1 Macro

0.702

Low accuracy expected:

- Only 2 000 training samples
- Vanishing gradients on ~150-token sequences
- Fluctuating loss curve confirms instability

Bidirectional LSTM — Architecture & Results

2 000 training samples · 5 epochs · CPU quick mode

LSTM Gates (solving vanishing gradients)

Forget

$f_t = \sigma(W_f \cdot [h_{t-1}, x_t])$

What to erase from memory

Input

$i_t = \sigma(W_i \cdot [h_{t-1}, x_t])$

What new info to write

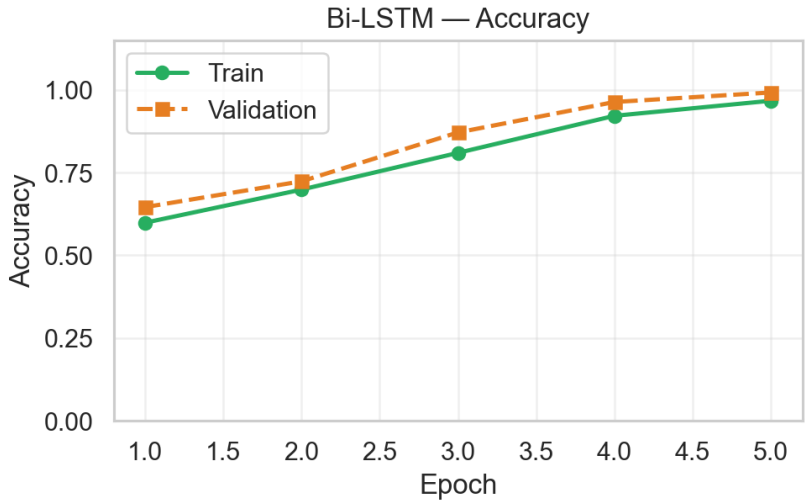
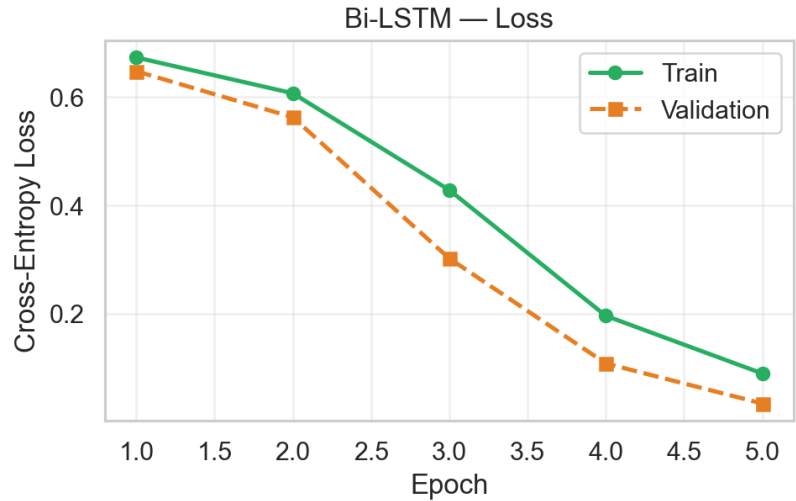
Output

$o_t = \sigma(W_o \cdot [h_{t-1}, x_t])$

What to expose as h_t

Bidirectional: $h = [\text{forward_hT} ; \text{backward_h}_1] \rightarrow \text{full context}$

Bi-LSTM Training History (Quick Mode, 5 Epochs)



Test Results

Accuracy

99.04 %

F1 Macro

0.9904

Misclassified

43 / 4 489

Trained on just 2 000 samples — same accuracy as Random Forest (36k samples).

Gate mechanism eliminates vanishing gradients; steady convergence over 5 epochs.

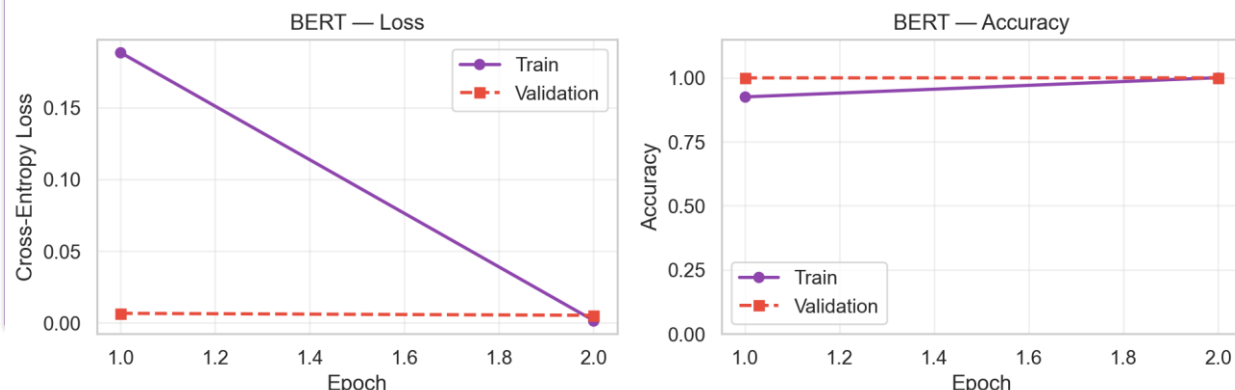
BERT — Transformer Fine-tuning

bert-base-uncased · 2 000 training samples · 2 epochs

Architecture

- ▶ 12 Transformer layers, 768 hidden dims, 12 attention heads
- ▶ Pre-trained on BookCorpus + Wikipedia (3.3B words)
- ▶ Self-attention: every token attends to every other
- ▶ Fine-tuned end-to-end with AdamW ($\text{lr} = 2 \times 10^{-5}$)
- ▶ Classification head on [CLS] token → Linear(768, 2)

BERT Training History (Quick Mode, 2 Epochs)



Test Results

Accuracy

99.87 %

F1 Macro

0.9987

Misclassified

6 / 4 489

Epochs

2

After just 1 epoch: val acc = 99.91%

After 2 epochs: train loss → 0.0017

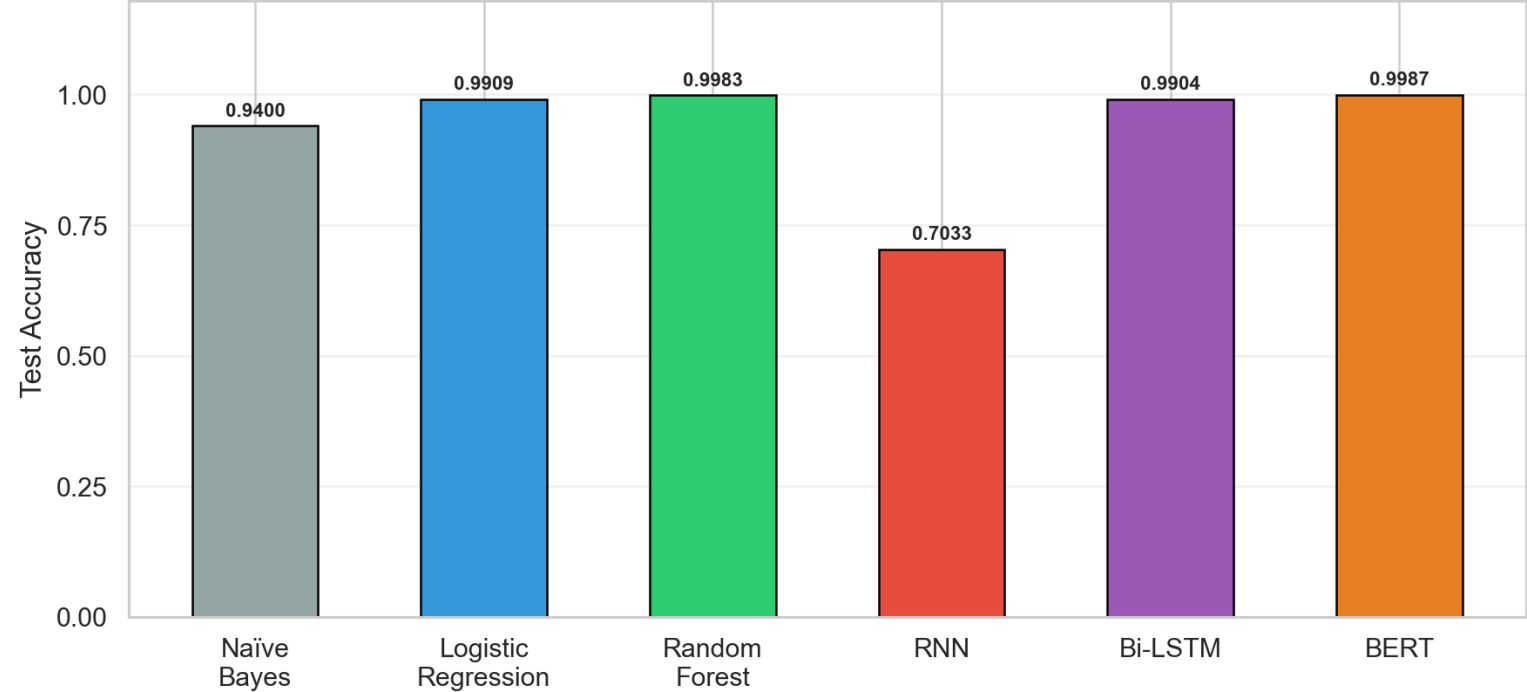
BERT's pre-trained representations need minimal fine-tuning — the [CLS] token learns near-perfect linear separation in 2 epochs.

Best result across all 6 models.

Model Comparison

All results on the ISOT test set

All Models — Test Accuracy Comparison



Summary Table

Model	Accuracy
Naive Bayes	94.0 %
Logistic Regression	99.1 %
Random Forest	99.8 %
RNN	70.3 %
Bi-LSTM	99.0 %
BERT	99.87 %

Conclusions

Key Findings

- ▶ ISOT corpus has strong stylistic signals — even Naive Bayes hits 94 %
- ▶ Random Forest (TF-IDF) achieves 99.8 % — best classical result
- ▶ Vanilla RNN struggles: vanishing gradients on 150-token sequences → 70.3 %
- ▶ Bi-LSTM solves this via gates: 99.04 % accuracy from only 2 000 samples
- ▶ BERT achieves 99.87 % — the best result — in just 2 epochs of fine-tuning
- ▶ Architecture quality > dataset size when the signal-to-noise ratio is high